sealed.

PROJECT REPORT

# A buyer-controlled AI assistant for car-dealer negotiation.

Finding comparable offers.
Coordinating the conversation.
Keeping the final decision with the buyer.

Written evidence | Private ceiling | Buyer approval

Problem statement, objectives, proposed solution, user journey, implementation outcomes and future scope.

October 18, 2026

Project stage: working local prototype

# Contents

A project overview for reviewers, collaborators and prospective users.


Executive summary 3

1. Introduction and background 3

2. Problem statement 4

3. Aim and objectives 4

4. Intended users and project scope 5

5. Existing approaches and the remaining gap 5

6. Proposed solution and guiding principles 6

7. User journey and illustrative scenario 6

8. Main features and user experience 8

9. Development approach and system overview 8

10. Implementation status and evaluation 9

11. Expected benefits and measures of success 9

12. Limitations and practical considerations 10

13. Future scope and development priorities 10

14. Conclusion 11

15. References and evidence basis 11


> This report distinguishes the problem being addressed, the intended benefits, the features implemented and the results actually demonstrated. Illustrative examples are not claims of completed vehicle purchases.

# Executive summary

Purchasing a car involves more than finding an attractive listing. A buyer must identify suitable dealers, request comparable prices, clarify excluded fees, follow up on unanswered messages and negotiate without losing control of the decision. Managing these conversations separately can make the process demanding, especially for buyers with limited time or negotiation experience.

Sealed is a project designed to support this process through an AI-assisted negotiation workflow. The buyer describes the required car, chooses a public target price and sets a private maximum acceptable price. The system helps find dealers, requests written quotes, organizes their responses and carries out a bounded negotiation. The buyer reviews the proposed deal and must approve its exact terms before the system requests purchase paperwork.

The central design principle is that the buyer's private maximum should not become a seller-facing negotiation signal. Sealed separates communication from decision authority: Fo handles authorized communication, while application rules control permitted facts, prices and actions. Written evidence supports quote comparison, and the buyer retains the final decision.

A working local prototype has been implemented, including dealer discovery, email outreach, quote extraction, call tracking, a dashboard and approval-based completion. Automated and controlled demonstrations validate the main workflow. Some individual integrations have also been exercised live. The complete live journey and the latest call protocol still require supervised acceptance. The project has not yet measured savings or time reduction in a real buyer study.

# 1. Introduction and background

Car buyers can research vehicles online, yet collecting and negotiating reliable dealer offers remains a separate task. A displayed vehicle price may differ from the amount payable after taxes, registration, dealer fees or additional products. A useful comparison therefore requires a written out-the-door total for the specified vehicle and clearly stated terms.

Digital convenience and buyer control are relevant needs. Cox Automotive's January 2026 publication of its 2025 Car Buyer Journey Study reported that 37% of buyers wanted to finalize the purchase price online, while 19% actually did so. The study surveyed 2,300 recent new- and used-vehicle buyers. These results describe a gap between preferred and actual purchase processes; they do not measure demand for Sealed specifically. Source 1

A separate Cox Automotive study published in August 2025 identified transparency, feeling informed, supportive staff and control as important contributors to shopper satisfaction. It also described waiting and finalizing terms among stressful parts of the journey. This supports the project's emphasis on clear offers and buyer approval. Source 2

Sealed focuses on coordinating quote collection and negotiation before a buyer commits. Test drives, inspections and the final transaction can still involve direct interaction with the dealership.

# 2. Problem statement

> **Car buyers lack a simple, buyer-controlled way to collect comparable written offers from multiple dealers, manage follow-ups and negotiate toward a chosen price while keeping their maximum willingness to pay private.**

The problem has several connected parts:

| Challenge | Effect on the buyer | Project response |
|---|---|---|
| Finding suitable sellers | The buyer may not know which nearby dealers to approach | Suggest dealers from public sources and let the buyer choose |
| Inconsistent quote formats | A listing price, monthly payment and all-in total cannot be compared directly | Request written out-the-door totals and flag incomplete terms |
| Repeated follow-ups | Separate emails and calls require attention and tracking | Coordinate authorized messages and bounded call nudges |
| Negotiation uncertainty | The buyer may struggle to decide what to say or which price to cite | Use a fixed strategy based on the public target and actual written quotes |
| Private budget exposure | Revealing the maximum can weaken the buyer's negotiating position | Keep the ceiling separate from external communication |
| Trust in automation | An assistant could overstate facts or act beyond the buyer's intent | Check communication, preserve evidence and require exact-term approval |

These are the design problems addressed by the prototype. Their frequency and severity for Sealed's intended users still require direct user research.

# 3. Aim and objectives

The aim is to make multi-dealer quote collection and negotiation easier to manage while preserving the buyer's privacy, understanding and authority.

The project objectives are to:

1. Accept a clear vehicle requirement, location, timeline and price preference.
2. Support buyers who already know dealers and buyers who need help finding them.
3. Collect written totals with enough context for meaningful comparison.
4. Reduce manual coordination through authorized email and phone communication.
5. Negotiate using truthful, supported prices within a fixed round plan.
6. Keep the private ceiling outside Fo, seller communication and quote-extraction model inputs.
7. Explain progress, incomplete evidence and review needs through one dashboard.

8. Require approval before requesting purchase paperwork and written confirmation before displaying completion.
9. Preserve progress across interruptions without blindly repeating outreach.

The desired outcome is an informed decision supported by documented offers. Achieving the target price depends on seller responses and is not guaranteed.

# 4. Intended users and project scope

### Intended users

The initial users are individual car buyers who know their approximate vehicle requirements but want help coordinating dealer conversations. Examples include first-time buyers, students and recent graduates, busy professionals, and buyers who prefer to compare written terms before visiting a dealership. These groups are proposed users; the project has not yet validated adoption within them.

Dealership staff are the communication counterpart. Clear specifications and genuine purchase intent can help them respond with relevant written terms.

### Current scope

The prototype supports one vehicle requirement, up to five sellers, a US ZIP-based dealer search of approximately 20 miles, email and authorized calls, and exactly two negotiation rounds. A user can enter sellers manually or choose researched candidates. The application currently runs locally.

The project does not execute payments, complete financing, inspect vehicles, verify legal title or independently certify inventory. For used cars, mileage, condition, history and equipment can make offers materially different even when the model and year match. The buyer must review those differences; an apparently lower total alone does not establish the better vehicle.

# 5. Existing approaches and the remaining gap

Buyers can already combine several approaches. The following comparison concerns types of workflow, not a claim that every product in a category behaves alike.

| Approach | What it helps with | Work that can remain for the buyer |
|---|---|---|
| Listing and search websites | Discovering vehicles and advertised prices | Confirming availability, obtaining final totals and following up |
| Direct dealer email and phone calls | Obtaining information from the seller | Tracking multiple conversations and negotiating consistently |
| Personal spreadsheets and notes | Recording quotes in one place | Entering terms, checking evidence and taking every next action |
| General AI assistance | Explaining concepts and drafting messages | Connecting communication to verified evidence and approval controls |

| Approach | What it helps with | Work that can remain for the buyer |
|---|---|---|
| Sealed's proposed workflow | Joining discovery, written offers, bounded communication and buyer approval | Reviewing exceptions, assessing the car and deciding whether to proceed |

The project opportunity is to bring these coordination steps together. Its value proposition is a manageable, documented buying process with explicit limits on what the assistant may decide and communicate. Broad superiority over existing services has not been established through comparative testing.

# 6. Proposed solution and guiding principles

Sealed provides a single workspace for the buyer's requirements, selected dealers, messages, quotes, negotiation status and final decision. An external assistant, Fo, performs authorized communication. Sealed supplies the permitted public facts and checks the proposed communication before allowing the next action.

Two price concepts are deliberately separated:

- **Public target:** the price the buyer would like to obtain. This may be used in a counteroffer.
- **Private ceiling:** the maximum used inside the backend when deciding whether a written offer is eligible to be proposed. It is not sent to Fo or sellers.

The system follows four guiding principles. Communication should use supported facts; comparable prices should come from written seller evidence; automation should have bounded actions; and a purchase decision should remain with the buyer. A call's spoken price is not treated as a firm written offer.

Email wording is generated within these limits rather than prescribed word for word. Code checks routing, required details, allowed prices and commitments. Calls similarly require a checked plan containing the buyer identity, number, opening and task before a separate dialing permission is issued.

These checks reduce tested failure modes. They do not prove every possible natural-language statement correct or mechanically restrict all of Fo's tools.

# 7. User journey and illustrative scenario




The buyer authorizes contact and approves the final proposal. The private ceiling is excluded from external communication.

### Stage 1: Describe the purchase

The buyer enters the car specification, ZIP, deadline, public target and private ceiling. More precise requirements improve the relevance of seller responses. Creating the negotiation alone contacts nobody.

### Stage 2: Choose dealers

The buyer adds known dealers or authorizes public dealer research. Suggested contacts are checked against published sources, and the buyer selects whom to approach. Discovery is distinct from permission to contact sellers.

### Stage 3: Collect written offers

After authorization, Fo asks for written out-the-door totals. Incomplete replies can receive one clarification request. An eligible silent seller can receive an authorized call asking for a written response.

### Stage 4: Compare and negotiate

The system evaluates supported written terms. Higher-priced sellers may be told the lowest genuine competing written total; a seller holding the lowest quote may receive a counter at the buyer's public target. The fixed plan does not increase offers toward the private ceiling.

### Stage 5: Review the proposal

The buyer sees the seller, amount, delivery terms, supporting evidence and flags. The buyer can approve the exact proposal, reject it or cancel. Eligibility within the private ceiling does not automatically authorize the purchase.

### Stage 6: Obtain written confirmation

Approval permits a request for paperwork at the agreed terms. The system waits for the seller's matching written confirmation before showing completion and a receipt. No payment is made through this workflow.

### Illustrative buyer scenario

A buyer seeks a Honda CR-V around model year 2015 and chooses a public target of $10,000. The buyer privately sets a maximum and selects several dealers. One seller replies with an asking price without fees, another provides a complete written total, and a third does not respond.

Sealed requests clarification from the first seller, retains the supported offer from the second and can use an authorized call nudge for the third. It negotiates only after appropriate written evidence exists. At the end of the rounds, it shows a qualifying offer or explains that review is needed. The buyer still checks mileage, condition, vehicle history and suitability before proceeding.

This scenario explains intended behavior. It is not a record of a completed Honda purchase or evidence that a dealer accepted $10,000.

# 8. Main features and user experience

| Feature | Purpose for the user |
|---|---|
| Requirements and pricing form | Describe the purchase once and distinguish target from private maximum |
| Dealer discovery and manual entry | Start even when the buyer has no existing dealer list |
| Written quote collection | Obtain a clearer basis for comparison than an advertised price alone |
| Quote clarification | Identify missing fees, specification details or other unsupported terms |
| Bounded negotiation | Carry out consistent, truthful counters and price-match requests |
| Call progress and outcomes | Show when a call result is pending and what Fo later reports |
| Dealer history | Review the messages, replies, decisions and evidence for each seller |
| Alerts and review states | Make uncertainty visible instead of silently continuing |
| Proposal approval and rejection | Preserve the buyer's control over the final terms |
| Final receipt and fee comparison | Summarize the written outcome and available itemized costs |

The dashboard uses a pinned summary and a next-step card to explain what is happening. Overview, Dealers, Activity and Receipt tabs organize the details. The private ceiling can be displayed through a separate endpoint and is kept out of event history and browser storage. Backend restart can require the buyer to restore that value before new actions continue.

# 9. Development approach and system overview

Development proceeded in stages: define the buying problem and trust boundaries; build the negotiation rules; connect controlled communication; validate written replies; add durable processing and a background worker; connect the dashboard; then improve dealer discovery, review handling and call progress.

The application combines a React dashboard with a Python/FastAPI backend. SQLite holds public job progress, evidence references and pending actions. Fo supplies communication and public research through MCP. Gmail supplies written message evidence. A separate Claude extraction step turns authorized seller text into structured terms, which application code validates.

The key division of responsibility is organizational as well as technical: the buyer defines the task and approves a proposal; Sealed controls the permitted workflow; Fo communicates within the authorized task; and written seller evidence supports the comparison. The private ceiling remains in backend memory.

The team integrated Renu's dashboard contributions and extended them with clearer discovery states, call progress, final outcomes and a corrected review indicator. Implementation details and module-level evidence are documented separately in the technical progress report.

# 10. Implementation status and evaluation

The current deliverable is a working local prototype. Its major workflow stages are implemented, but implementation, synthetic validation and live acceptance represent different levels of evidence.

| Area | Evidence available | Interpretation |
|---|---|---|
| Core negotiation and approval | Complete synthetic runs with written confirmation and replay checks | The implemented workflow can reach a controlled final outcome |
| Email communication | Controlled live sends and a target-price counter checked against Gmail evidence | Individual outreach and auditing paths have worked live |
| Quote extraction | One authorized live sample, plus automated validation tests | Missing specification evidence correctly prevents a firm result |
| Dealer research | Historical live research and checked candidates, plus simulated discovery tests | Candidate discovery works; stock and fresh availability remain unconfirmed |
| Call lifecycle | Automated checks and synthetic browser updates | The new checked-plan protocol still needs a fresh consenting live call |
| Dashboard | Browser checks of progress, activity, approval and receipt views | The tested interface displays the expected state transitions |

The recorded automated checks include 586 passing backend tests, 9 passing frontend tests, a successful production build and lint checks. Browser testing confirmed pending call display, a final reported no-answer outcome, activity updates and a synthetic final receipt. Tested mobile views had no horizontal overflow.

A controlled synthetic negotiation used a $100 written quote and $90 public target, followed by explicit simulated approval and written confirmation. Those sample values demonstrate system behavior. They are not measured car-purchase savings or proof of market performance.

The evaluation has not yet included a representative buyer study, an independent audit of all Fo actions, or a complete new live dashboard journey from discovery to the seller's final written confirmation.

# 11. Expected benefits and measures of success

Sealed is intended to reduce repetitive coordination, improve the readability of offers and give the buyer a clearer decision trail. Keeping the private ceiling out of seller communication is intended to preserve negotiating discretion. Requiring written evidence and approval is intended to improve confidence in what the system presents.

These are expected benefits. Real time savings, purchase savings, adoption and satisfaction have not yet been quantified for this project.

| Evaluation question | Proposed measure |
|---|---|
| Does it reduce manual effort? | Buyer time spent and number of manual follow-ups versus the buyer's normal process |
| Does it improve quote quality? | Share of responses with complete, comparable written terms |

| Evaluation question | Proposed measure |
| --- | --- |
| Is communication reliable? | Correct-recipient deliveries, duplicate actions, unresolved attempts and successful recovery |
| Does the buyer understand the result? | Ability to identify the seller, total, conditions and reason for a review flag |
| Does the privacy design hold? | Observed ceiling disclosures and unauthorized price or commitment statements in tested communications |
| Does it improve the purchase outcome? | Final confirmed terms compared with an appropriate documented baseline for the same purchase |

Both successful and unsuccessful negotiations should be included in future evaluation. Savings comparisons must account for changes in vehicle condition, fees and terms rather than comparing unrelated advertised prices.

# 12. Limitations and practical considerations

Seller participation is outside the application's control. Some dealers may not reply, may prefer direct contact, or may decline the requested price. The two-round plan can finish without a qualifying offer.

Discovery depends on public information. Website access restrictions can exclude valid dealers, and an accessible listing does not prove stock. The 20-mile filter uses approximate ZIP-centroid distances rather than driving distance.

Written terms can be incomplete or ambiguous. Unsupported extraction is held for review. Phone-only sellers also need a trusted written identity handoff before an unknown email can count as authenticated dealer evidence.

An earlier live call exposed a buyer-name mismatch and an unclear progress response. The new checked call plan and result lifecycle address those tested failure modes, but await fresh live acceptance. Identity and follow-up checks rely on Fo-reported evidence; they are not independent voice or scheduler audits.

The current application is local and has no multi-user sign-in system. Hosted operation would require additional account isolation and deployment work. A backend restart clears the private ceiling. Cancelling a negotiation stops queued application actions but cannot recall an external call already underway.

Finally, communication checks and model extraction do not replace the buyer's assessment of the vehicle, financing, contract or final purchase paperwork.

# 13. Future scope and development priorities

### Immediate validation

The next priority is a consenting one-call acceptance that confirms the correct buyer identity, a single dialing permission, visible progress and a usable final report. This should be followed by a complete controlled live negotiation through written quote, counter, buyer approval and written confirmation. Restart and review handling should be exercised during that acceptance.

### Product improvements

Further work can improve guided vehicle requirements, explain why a quote is incomplete, make review actions easier to understand and establish a clearer trusted-contact onboarding process for phone-only dealers. User interviews and observed tasks should inform those changes.

### Broader deployment and evaluation

A hosted version would need user authentication, per-account separation and operational recovery appropriate to real users. A pilot should measure effort, quote completeness and decision confidence before making claims about savings. Service costs and support requirements should also be measured before selecting a pricing model; the project currently has no validated revenue model.

### Longer-term possibilities

The workflow may later apply to other purchases where specifications and written offers can be compared meaningfully. This would require domain-specific rules. Vehicle history, inspection workflows, richer location filtering and more detailed cost comparison are possible extensions rather than completed features.

---

# 14. Conclusion

Sealed addresses the coordination work between deciding what car to buy and receiving a clear, acceptable written offer. It combines dealer selection, authorized communication, evidence-based quote comparison and a bounded negotiation process while preserving the buyer's final approval.

The project demonstrates a working local workflow and several successful individual live integration checks. Its strongest contribution is the combination of a private-ceiling boundary, supported negotiation facts, visible progress and explicit approval. Completing controlled live acceptance and testing with buyers are the next steps toward establishing its practical value.

---

# 15. References and evidence basis


1. Cox Automotive. *Car Buyer Journey Study Finds Efficiency, Digital Tools and AI Drive Record Satisfaction.* January 13, 2026; reporting research conducted in fall 2025. Read the study announcement.
2. Cox Automotive. *Key Insights into Successful Car Deals: Drivers of Car Shopping Satisfaction.* August 26, 2025. Read the study announcement.
3. Sealed project. *Implementation and validation report.* October 18, 2026. Repository document: `docs/progress_report_2026-10-18.md`.
4. Sealed project. Architecture, dashboard, quote parsing, dealer discovery and call lifecycle documentation; current source and recorded test evidence at implementation snapshot `941a591`.


External studies provide background context. Project results derive from the repository and recorded controlled tests. The persona and Honda example are illustrative; proposed benefits and future features are identified as such.